# Parametric Survival Models

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## Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# printr is not loaded here: it inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
```

```
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
# Summarizes key information about statistical objects in tidy tibbles
library(broom)
# Provides suite of functions to work with regression model
# `broom::tidy()` tibbles
library(broom.helpers)
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")
    )
)

knitr::opts_chunk$set(message = FALSE)
options(digits = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
# use the `filter()` function from dplyr() by default
conflicts_prefer(dplyr::filter)
legend_text_size <- 9
run_graphs <- TRUE
```

# 1 Parametric Survival Models

Video lecture

David Rocke's Epi 204 lecture *Parametric Survival Regression*[a] (slides[b]) covers exponential and Weibull models, accelerated failure time models, the `survreg()` function, and complementary log-log plots, using the `anderson` data [1].

[a] https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/EPI204-2021-05-20.mp4
[b] https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/Lecture%20Parametric%20Survival%20Models.pdf

## 1.1 Exponential Distribution

The exponential distribution[1] is the baseline parametric model in survival analysis. It assumes a constant hazard rate[2] $\lambda > 0$ over time. This constant hazard implies that the event probability in any short interval depends only on the interval duration and not on elapsed time (the memoryless property).

[1] https://morrison-lab.github.io/pds/random-variables.html#def-exponential
[2] intro-to-survival-analysis.qmd#def-hazard

$$
\begin{aligned}
\mathrm{f}(t) &\stackrel{\text{def}}{=} \lambda \mathrm{e}^{-\lambda t} && \text{(probability density function definition for } t \geq 0) \\
\log\{\mathrm{f}(t)\} &= \log\{\lambda\} - \lambda t && \text{(taking natural logarithm of density)} \\
\mathrm{F}(t) &\stackrel{\text{def}}{=} \int_0^t \mathrm{f}(u)du && \text{(cumulative distribution function definition)} \\
&= \int_0^t \lambda \mathrm{e}^{-\lambda u} du && \text{(substituting probability density function)} \\
&= \left[-\mathrm{e}^{-\lambda u}\right]_0^t && \text{(evaluating antiderivative)} \\
&= 1 - \mathrm{e}^{-\lambda t} && \text{(evaluating limits of integration)} \\
\mathrm{S}(t) &\stackrel{\text{def}}{=} 1 - \mathrm{F}(t) && \text{(survival function definition)} \\
&= 1 - (1 - \mathrm{e}^{-\lambda t}) && \text{(substituting cumulative distribution function)} \\
&= \mathrm{e}^{-\lambda t} && \text{(simplifying terms)} \\
\Lambda(t) &\stackrel{\text{def}}{=} -\log\{\mathrm{S}(t)\} && \text{(cumulative hazard function definition)} \\
&= -\log\{\mathrm{e}^{-\lambda t}\} && \text{(substituting survival function)} \\
&= \lambda t && \text{(simplifying logarithm of exponential)} \\
\lambda(t) &\stackrel{\text{def}}{=} \frac{\partial}{\partial t}\Lambda(t) && \text{(hazard function definition as derivative of cumulative hazard)} \\
&= \frac{\partial}{\partial t}(\lambda t) && \text{(substituting cumulative hazard)} \\
&= \lambda && \text{(differentiating linear term)} \\
\mathbb{E}[T] &\stackrel{\text{def}}{=} \int_0^\infty \mathrm{S}(t)dt && \text{(expectation formula for non-negative continuous random variable)} \\
&= \int_0^\infty \mathrm{e}^{-\lambda t} dt && \text{(substituting survival function)} \\
&= \left[-\frac{1}{\lambda}\mathrm{e}^{-\lambda t}\right]_0^\infty && \text{(evaluating antiderivative)} \\
&= \lambda^{-1} && \text{(evaluating limits of integration)}
\end{aligned}
$$

**Theorem 1.1** (The exponential distribution is memoryless)**.** *If $T$ has an exponential distribution with rate $\lambda > 0$, then for all $s \geq 0$ and $t \geq 0$,*

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t).$$

Proof

*Proof.*

$$
\begin{aligned}
\mathbb{P}(T > s + t \mid T > s) &= \frac{\mathbb{P}(T > s + t,\ T > s)}{\mathbb{P}(T > s)} && \text{(definition of conditional probability)} \\
&= \frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)} && \text{(the event } T > s + t \text{ implies } T > s) \\
&= \frac{\mathrm{S}(s + t)}{\mathrm{S}(s)} && \text{(definition of the survival function)} \\
&= \frac{\mathrm{e}^{-\lambda(s+t)}}{\mathrm{e}^{-\lambda s}} && \text{(exponential survival function)} \\
&= \mathrm{e}^{-\lambda(s+t)+\lambda s} && \text{(quotient rule for exponentials)} \\
&= \mathrm{e}^{-\lambda t} && \text{(cancel } \lambda s) \\
&= \mathbb{P}(T > t) && \text{(exponential survival function)}
\end{aligned}
$$

□

**Example 1.1** (Remaining survival after 10 event-free weeks)**.** Let $T$ be exponential with rate $\lambda = 0.1$ events per week. For a subject who has been event-free for $s = 10$ weeks, the probability of lasting at least $t = 5$ more weeks is:

$$\begin{aligned}
\mathbb{P}(T > 15 \mid T > 10) &= \frac{\mathrm{e}^{-0.1\cdot 15}}{\mathrm{e}^{-0.1\cdot 10}} && \text{(ratio of survival functions)} \\
&= \frac{\mathrm{e}^{-1.5}}{\mathrm{e}^{-1}} && \text{(multiply)} \\
&= \mathrm{e}^{-0.5} && \text{(quotient rule for exponentials)} \\
&\approx 0.6065 && \text{(evaluate)}
\end{aligned}$$

This value equals $\mathbb{P}(T > 5) = \mathrm{e}^{-0.1\cdot 5} = \mathrm{e}^{-0.5}$, the probability that a new subject lasts 5 weeks: the 10 weeks already survived do not change the outlook.

**Example 1.2** (A Weibull distribution with shape 2 is not memoryless (counterexample))**.** Now let $T$ be Weibull with $\lambda = 0.1$ and $p = 2$, so $\mathrm{S}(t) = \mathrm{e}^{-(0.1t)^2}$ and the hazard increases over time. Then:

$$\begin{aligned}
\mathbb{P}(T > 15 \mid T > 10) &= \frac{\mathrm{e}^{-(0.1\cdot 15)^2}}{\mathrm{e}^{-(0.1\cdot 10)^2}} && \text{(ratio of survival functions)} \\
&= \frac{\mathrm{e}^{-(1.5)^2}}{\mathrm{e}^{-(1)^2}} && \text{(multiply)} \\
&= \frac{\mathrm{e}^{-2.25}}{\mathrm{e}^{-1}} && \text{(square)} \\
&= \mathrm{e}^{-1.25} && \text{(quotient rule for exponentials)} \\
&\approx 0.2865 && \text{(evaluate)}
\end{aligned}$$

whereas $\mathbb{P}(T > 5) = \mathrm{e}^{-(0.5)^2} = \mathrm{e}^{-0.25} \approx 0.7788$. A subject who has already lasted 10 weeks is less likely to last 5 more weeks than a new subject is, because the hazard has risen in the meantime.

## 1.2 Weibull Distribution

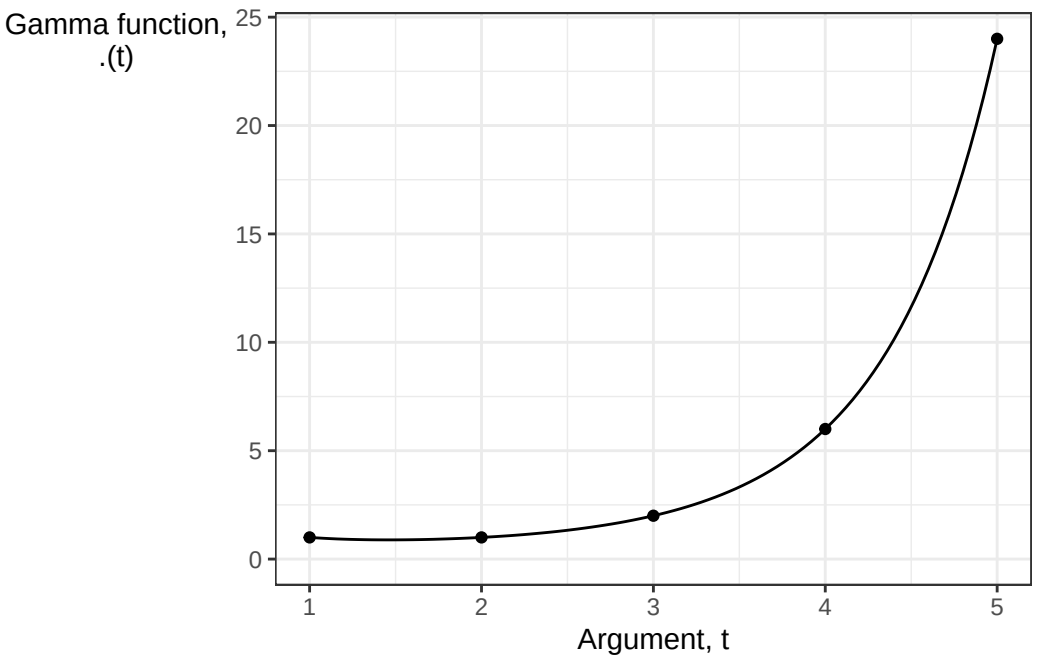
The Weibull distribution[3] generalizes the exponential distribution by introducing a shape parameter $p > 0$ alongside the scale parameter $\lambda > 0$ [2]. This flexibility allows the hazard rate to change monotonically over time, accommodating situations where risk increases or decreases as time elapses.

Using the parameterization of Kalbfleisch and Prentice [2]:

[3] https://morrison-lab.github.io/pds/distributions.html#def-weibull

$$
\begin{aligned}
\mathrm{f}(t) &\stackrel{\text{def}}{=} \lambda p(\lambda t)^{p-1}\mathrm{e}^{-(\lambda t)^p} && \text{(probability density function for } t \geq 0\text{)} \\
\mathrm{F}(t) &\stackrel{\text{def}}{=} \int_0^t \mathrm{f}(u)du && \text{(cumulative distribution function definition)} \\
&= \int_0^t \lambda p(\lambda u)^{p-1}\mathrm{e}^{-(\lambda u)^p}du && \text{(substituting density function)} \\
&= \left[-\mathrm{e}^{-(\lambda u)^p}\right]_0^t && \text{(evaluating antiderivative via substitution } w = (\lambda u)^p\text{)} \\
&= 1 - \mathrm{e}^{-(\lambda t)^p} && \text{(evaluating limits of integration)} \\
\mathrm{S}(t) &\stackrel{\text{def}}{=} 1 - \mathrm{F}(t) && \text{(survival function definition)} \\
&= 1 - \left(1 - \mathrm{e}^{-(\lambda t)^p}\right) && \text{(substituting cumulative distribution function)} \\
&= \mathrm{e}^{-(\lambda t)^p} && \text{(simplifying terms)} \\
\Lambda(t) &\stackrel{\text{def}}{=} -\log\{\mathrm{S}(t)\} && \text{(cumulative hazard function definition)} \\
&= -\log\left\{\mathrm{e}^{-(\lambda t)^p}\right\} && \text{(substituting survival function)} \\
&= (\lambda t)^p && \text{(simplifying logarithm of exponential)} \\
\log\{\Lambda(t)\} &= \log\{(\lambda t)^p\} && \text{(taking natural logarithm of cumulative hazard)} \\
&= p\log\{\lambda t\} && \text{(applying exponent rule for logarithms)} \\
&= p\log\{\lambda\} + p\log\{t\} && \text{(applying product rule for logarithms)} \\
\lambda(t) &\stackrel{\text{def}}{=} \frac{\partial}{\partial t}\Lambda(t) && \text{(hazard function definition)} \\
&= \frac{\partial}{\partial t}((\lambda t)^p) && \text{(substituting cumulative hazard)} \\
&= p(\lambda t)^{p-1} \cdot \lambda && \text{(applying power and chain rules of differentiation)} \\
&= \lambda p(\lambda t)^{p-1} && \text{(rearranging factors)} \\
\mathbb{E}[T] &\stackrel{\text{def}}{=} \int_0^\infty \mathrm{S}(t)dt && \text{(expectation formula for non-negative random variable)} \\
&= \int_0^\infty \mathrm{e}^{-(\lambda t)^p}dt && \text{(substituting survival function)} \\
&= \frac{1}{\lambda p}\int_0^\infty u^{\frac{1}{p}-1}\mathrm{e}^{-u}du && \text{(substituting } u = (\lambda t)^p \implies t = \lambda^{-1}u^{1/p}\text{)} \\
&= \lambda^{-1} \cdot \Gamma\left(1 + \frac{1}{p}\right) && \text{(applying definition of Gamma function } \Gamma(z)\text{)}
\end{aligned}
$$

> **Note**
>
> Recall from calculus:
> - $\Gamma(t) \stackrel{\text{def}}{=} \int_{u=0}^\infty u^{t-1}\mathrm{e}^{-u}du$
> - $\Gamma(t) = (t-1)!$ for integers $t \in \mathbb{Z}$
> - It is implemented by the `gamma()` function in R.




Here are some Weibull density functions[4], with $\lambda = 1$ and $p$ varying:

```
library(ggplot2)
lambda <- 1
ggplot() +
  geom_function(
    aes(col = "0.25"),
    fun = \(x) dweibull(x, shape = 0.25, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "0.5"),
    fun = \(x) dweibull(x, shape = 0.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1"),
    fun = \(x) dweibull(x, shape = 1, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1.5"),
    fun = \(x) dweibull(x, shape = 1.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "2"),
    fun = \(x) dweibull(x, shape = 2, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "5"),
    fun = \(x) dweibull(x, shape = 5, scale = 1 / lambda)
  ) +
  theme_bw() +
  xlim(0, 2.5) +
  ylab("Density,\nf(t)") +
  theme(axis.title.y = element_text(angle = 0)) +
  theme(legend.position = "bottom") +
```

[4] https://morrison-lab.github.io/pds/random-variables.html#def-pdf

```
guides(
  col =
    guide_legend(
      title = "p",
      label.theme = element_text(size = 12)
    )
)
```

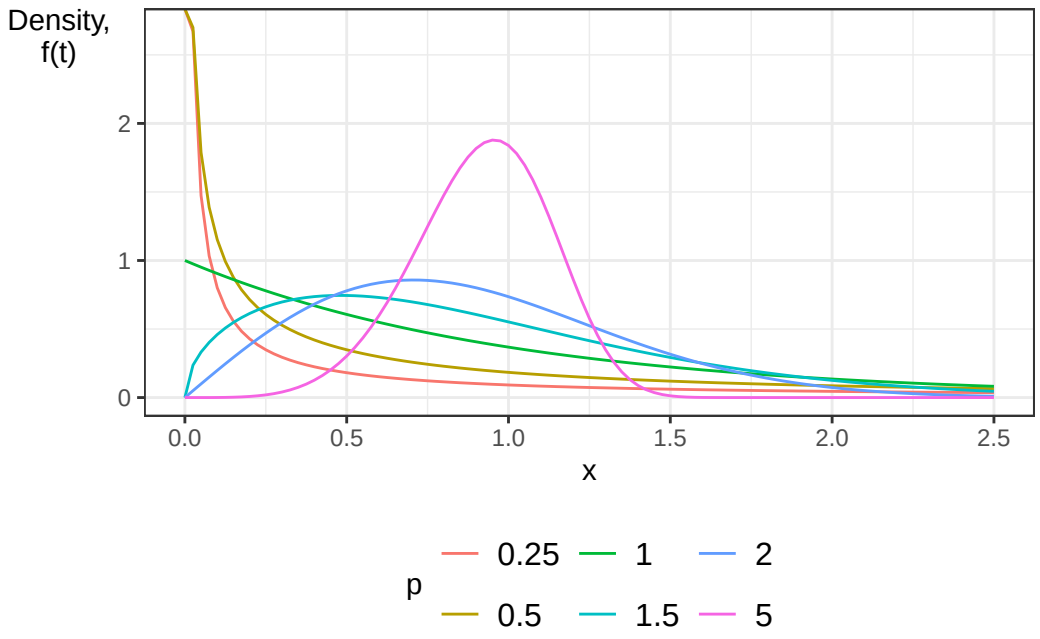



Figure 1: Density functions for Weibull distribution

### 1.2.1 Properties of Weibull hazard functions

**Theorem 1.2** (Properties of Weibull hazard functions)**.** *If $T$ has a Weibull distribution, then:*

- *When $p = 1$, the Weibull distribution simplifies to the exponential distribution.*
- *When $p > 1$, the hazard is strictly increasing: $\lambda'(t) > 0$.*
- *When $p < 1$, the hazard is strictly decreasing: $\lambda'(t) < 0$.*
- *$\log \Lambda(t)$ is a straight line relative to $\log t$: $\log \Lambda(t) = p \log \lambda + p \log t$.*

Proof

*Proof.* We prove each property of the Weibull hazard function $\lambda(t) = \lambda p(\lambda t)^{p-1}$:

1. **Simplification to Exponential distribution when $p = 1$:** Setting $p = 1$ in the Weibull hazard function yields:

$$\begin{aligned} \lambda(t) &= \lambda(1)(\lambda t)^{1-1} && \text{(substituting } p = 1 \text{ into Weibull hazard formula)} \\ &= \lambda \cdot 1 \cdot (\lambda t)^0 && \text{(simplifying exponent } 1 - 1 = 0) \\ &= \lambda && \text{(since } (\lambda t)^0 = 1 \text{ for } t > 0) \end{aligned}$$

This constant value is the hazard function[a] of the exponential distribution with parameter $\lambda$.

2. **Increasing hazard when $p > 1$**: Differentiating $\lambda(t) = \lambda^p p t^{p-1}$ with respect to $t$ gives:

$$\begin{aligned} \lambda'(t) &\overset{\text{def}}{=} \frac{\partial}{\partial t}(\lambda^p p t^{p-1}) && \text{(definition of hazard derivative)} \\ &= \lambda^p p(p-1)t^{p-2} && \text{(applying power rule of differentiation)} \end{aligned}$$

   Since $\lambda > 0$, $p > 0$, and $t > 0$, the sign of $\lambda'(t)$ depends entirely on the factor $(p-1)$. When $p > 1$, $(p-1) > 0$, so $\lambda'(t) > 0$ for all $t > 0$, meaning the hazard function is strictly increasing over time.
3. **Decreasing hazard when $p < 1$**: Using the derivative $\lambda'(t) = \lambda^p p(p-1)t^{p-2}$: When $p < 1$, $(p-1) < 0$, so $\lambda'(t) < 0$ for all $t > 0$, meaning the hazard function is strictly decreasing over time.
4. **Linear relationship between $\log \Lambda(t)$ and $\log t$**: Taking the natural logarithm[b] of the cumulative hazard function[c] $\Lambda(t) = (\lambda t)^p$:

$$\begin{aligned} \log\{\Lambda(t)\} &\overset{\text{def}}{=} \log\{(\lambda t)^p\} && \text{(substituting Weibull cumulative hazard)} \\ &= p\log\{\lambda t\} && \text{(applying logarithm exponent rule } \log(a^b) = b\log a) \\ &= p\log\{\lambda\} + p\log\{t\} && \text{(applying logarithm product rule } \log(ab) = \log a + \log b) \end{aligned}$$

   Defining $y \overset{\text{def}}{=} \log\{\Lambda(t)\}$ and $x \overset{\text{def}}{=} \log\{t\}$, this equation takes the linear form $y = a + bx$ with intercept $a = p\log\{\lambda\}$ and slope $b = p$.

□

[a] intro-to-survival-analysis.qmd#def-hazard
[b] https://morrison-lab.github.io/mds/algebra-exponentials.html#def-natural-log
[c] intro-to-survival-analysis.qmd#def-cuhaz

**Example 1.3** (Evaluating Weibull hazard shapes under different shape parameters)**.** Consider a clinical trial evaluating time to relapse in months following cancer treatment, modeled using a Weibull distribution with scale parameter $\lambda = 0.1$. We evaluate the hazard rate $\lambda(t)$ at months $t = 1, 6, 12$ under three distinct shape parameters:

1. **Decreasing hazard ($p = 0.5$)**: The hazard rate formula gives:

$$\lambda(t) = (0.1)(0.5)(0.1t)^{-0.5} = \frac{0.05}{\sqrt{0.1t}}$$

   - At $t = 1$ month: $\lambda(1) = \frac{0.05}{\sqrt{0.1}} \approx 0.1581$ events per month.
   - At $t = 6$ months: $\lambda(6) = \frac{0.05}{\sqrt{0.6}} \approx 0.0645$ events per month.
   - At $t = 12$ months: $\lambda(12) = \frac{0.05}{\sqrt{1.2}} \approx 0.0456$ events per month. The risk of relapse is highest immediately following treatment and declines over time.
2. **Constant hazard ($p = 1.0$)**: The hazard rate formula reduces to $\lambda(t) = \lambda = 0.1000$ events per month for all $t$. The risk of relapse remains constant regardless of time elapsed.
3. **Increasing hazard ($p = 2.0$)**: The hazard rate formula gives:

$$\lambda(t) = (0.1)(2.0)(0.1t)^{1.0} = 0.02t$$

   - At $t = 1$ month: $\lambda(1) = 0.02(1) = 0.0200$ events per month.
   - At $t = 6$ months: $\lambda(6) = 0.02(6) = 0.1200$ events per month.
   - At $t = 12$ months: $\lambda(12) = 0.02(12) = 0.2400$ events per month. The risk of relapse increases linearly over time.

**Exercise 1.1.** Verify the calculations in Example 1.3.

**Exercise 1.2.** In the setting of Example 1.3, explain how the shape parameter $p$ affects cumulative hazard[a] curves.

[a] intro-to-survival-analysis.qmd#def-cuhaz

The Weibull distribution provides more flexibility than the exponential. Figure 2 shows some Weibull hazard functions, with $\lambda = 1$ and $p$ varying:

```
library(ggplot2)
library(eha)
lambda <- 1

ggplot() +
  geom_function(
    aes(col = "0.25"),
    fun = \(x) hweibull(x, shape = 0.25, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "0.5"),
    fun = \(x) hweibull(x, shape = 0.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1"),
    fun = \(x) hweibull(x, shape = 1, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1.5"),
    fun = \(x) hweibull(x, shape = 1.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "2"),
    fun = \(x) hweibull(x, shape = 2, scale = 1 / lambda)
  ) +
  theme_bw() +
  xlim(0, 2.5) +
  ylab(expr(lambda)) +
  theme(axis.title.y = element_text(angle = 0)) +
  theme(legend.position = "bottom") +
  guides(
    col =
      guide_legend(
        title = "p",
        label.theme = element_text(size = 12)
      )
  )
```

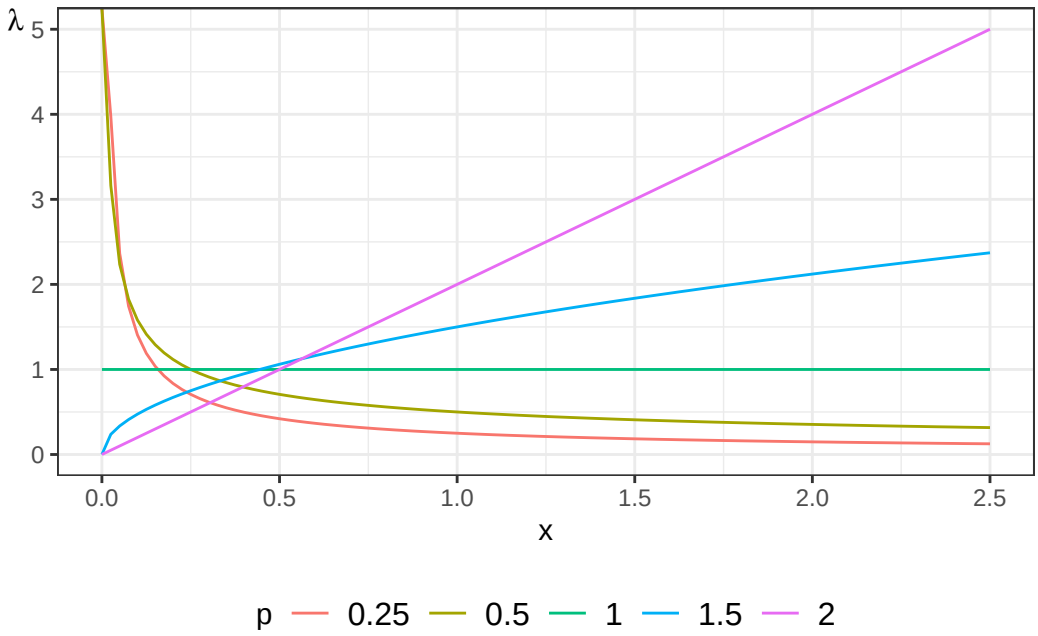



Figure 2: Hazard functions for Weibull distribution

---

```r
library(ggplot2)
lambda <- 1

ggplot() +
  geom_function(
    aes(col = "0.25"),
    fun = pweibull,
    args = list(shape = 0.25, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "0.5"),
    fun = pweibull,
    args = list(shape = 0.5, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "1"),
    fun = pweibull,
    args = list(shape = 1, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "1.5"),
    fun = pweibull,
    args = list(shape = 1.5, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "2"),
    fun = pweibull,
    args = list(shape = 2, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  theme_bw() +
  xlim(0, 2.5) +
  ylab("Survival\nprobability,\nS(t)") +
```

```r
theme(axis.title.y = element_text(angle = 0)) +
theme(legend.position = "bottom") +
guides(
  col =
    guide_legend(
      title = "p",
      label.theme = element_text(size = 12)
    )
)
```

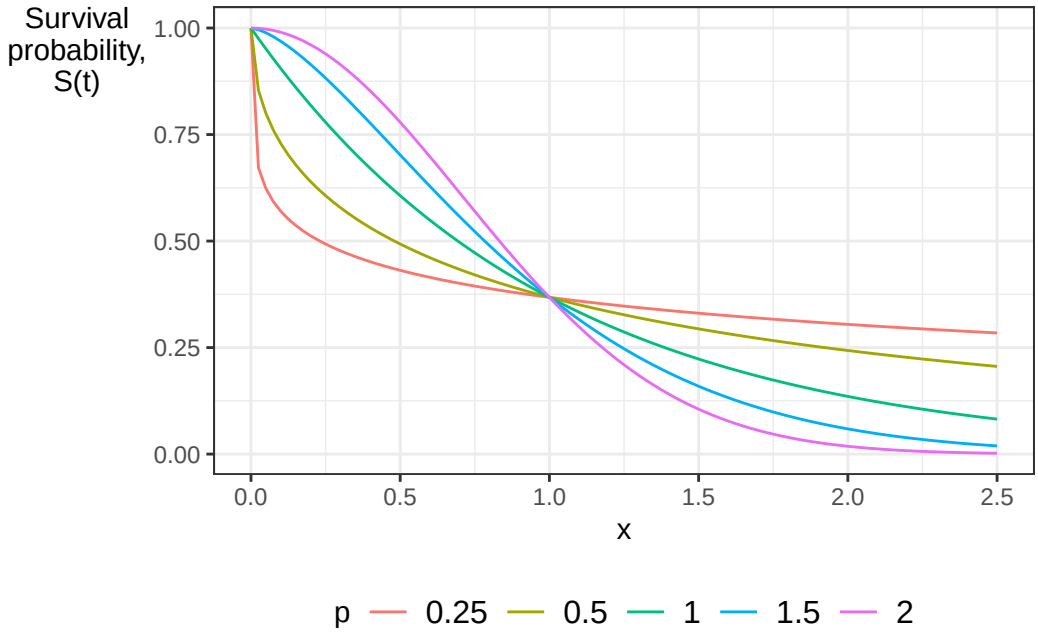



Figure 3: Survival functions for Weibull distribution

## 1.3 Exponential Regression

**Definition 1.1** (Exponential regression model)**.** An **exponential regression model** for an event time $T$ with predictor[a] vector $\underset{\sim}{x}$ assumes that, given $\underset{\sim}{x}$, $T$ has an exponential distribution[b] whose hazard rate[c] is the exponentiated linear predictor[d]:

$$\lambda(t \mid \underset{\sim}{x}) \overset{\text{def}}{=} \exp\{\eta(\underset{\sim}{x})\}, \quad \eta(\underset{\sim}{x}) \overset{\text{def}}{=} \underset{\sim}{x} \cdot \underset{\sim}{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p \tag{1}$$

[a] Intro-to-GLMs.qmd#def-predictors
[b] https://morrison-lab.github.io/pds/random-variables.html#def-exponential
[c] intro-to-survival-analysis.qmd#def-hazard
[d] Intro-to-GLMs.qmd#def-linear-predictor

**Example 1.4** (Exponential regression with one binary covariate)**.** Let $x_1 = 1$ for patients on a standard treatment and $x_1 = 0$ for patients on a new treatment, and suppose $\beta_0 = \log\{0.025\}$ and $\beta_1 = \log\{4\}$, with time measured in weeks. By Equation 1:

$$
\begin{aligned}
\lambda(t \mid x_1 = 0) &= \exp\{\beta_0\} && \text{(exponential regression hazard with } x_1 = 0) \\
&= 0.025 && (\exp\{\log\{0.025\}\} = 0.025) \\
\lambda(t \mid x_1 = 1) &= \exp\{\beta_0 + \beta_1\} && \text{(exponential regression hazard with } x_1 = 1) \\
&= \exp\{\beta_0\} \cdot \exp\{\beta_1\} && \text{(product rule for exponentials)} \\
&= 0.025 \cdot 4 && \text{(evaluate each factor)} \\
&= 0.1 && \text{(multiply)}
\end{aligned}
$$

Each group's mean event time is the reciprocal of its hazard rate, as derived in Section 1.1: $1/0.025 = 40$ weeks on the new treatment and $1/0.1 = 10$ weeks on the standard treatment.

Exponential regression models the hazard rate using a log link function[5]. At $\underset{\sim}{x} = \underset{\sim}{0}$, Equation 1 gives the baseline hazard rate:

$$
\begin{aligned}
\lambda_0(t) &\overset{\text{def}}{=} \lambda(t \mid \underset{\sim}{0}) && \text{(baseline hazard rate definition at } \underset{\sim}{x} = \underset{\sim}{0}) \\
&= \exp\{\eta(\underset{\sim}{0})\} && \text{(substituting } \underset{\sim}{x} = \underset{\sim}{0} \text{ into hazard function)} \\
&= \exp\{\beta_0 + (\beta_1 \cdot 0 + \cdots + \beta_p \cdot 0)\} && \text{(evaluating linear predictor at zero vector)} \\
&= \exp\{\beta_0\} && \text{(simplifying exponent)}
\end{aligned}
$$

When there are no additional predictors[6] ($\underset{\sim}{x} = \underset{\sim}{0}$), the baseline hazard[7] is constant: $\lambda = \exp\{\beta_0\}$. This model uses a log link as in a generalized linear model[8] for Poisson[9] or rate outcomes. Because the hazard rate does not depend on time $t$, the hazard ratio[10] comparing two covariate vectors $\underset{\sim}{x}$ and $\underset{\sim}{x}^*$ is constant over time:

$$
\begin{aligned}
{}_\lambda(t \mid \underset{\sim}{x} : \underset{\sim}{x}^*) &\overset{\text{def}}{=} \frac{\lambda(t \mid \underset{\sim}{x})}{\lambda(t \mid \underset{\sim}{x}^*)} && \text{(hazard ratio definition)} \\
&= \frac{\exp\{\eta(\underset{\sim}{x})\}}{\exp\{\eta(\underset{\sim}{x}^*)\}} && \text{(substituting exponential regression hazard rates)} \\
&= \exp\{\eta(\underset{\sim}{x}) - \eta(\underset{\sim}{x}^*)\} && \text{(applying quotient rule for exponentials)} \\
&= \exp\{(\underset{\sim}{x} - \underset{\sim}{x}^*) \cdot \underset{\sim}{\beta}\} && \text{(expressing difference in linear predictors)}
\end{aligned}
$$

Thus, exponential regression models possess proportional hazards[11] trivially.

## 1.4 Accelerated Failure Time

In a proportional hazards (PH) model[12], covariates act multiplicatively on the baseline hazard function[13]:

[5] Intro-to-GLMs.qmd#def-link-function
[6] Intro-to-GLMs.qmd#def-predictors
[7] proportional-hazards-models.qmd#def-base-hazard
[8] Intro-to-GLMs.qmd#def-glm
[9] https://morrison-lab.github.io/pds/distributions.html#def-poisson
[10] intro-to-survival-analysis.qmd#def-hazard-ratio
[11] proportional-hazards-models.qmd#def-pha
[12] proportional-hazards-models.qmd#def-ph-model
[13] proportional-hazards-models.qmd#def-base-hazard

$$
\begin{aligned}
\lambda(t \mid \underset{\sim}{x}) &\stackrel{\text{def}}{=} \lambda_0(t) \cdot \theta_\lambda(\underset{\sim}{x}) && \text{(proportional hazards hazard rate definition)} \\
\Lambda(t \mid \underset{\sim}{x}) &\stackrel{\text{def}}{=} \int_0^t \lambda(u \mid \underset{\sim}{x}) du && \text{(cumulative hazard function definition)} \\
&= \int_0^t \lambda_0(u) \cdot \theta_\lambda(\underset{\sim}{x}) du && \text{(substituting proportional hazard rate)} \\
&= \theta_\lambda(\underset{\sim}{x}) \cdot \Lambda_0(t) && \text{(factoring constant hazard multiplier out of integral)} \\
\mathrm{S}(t \mid \underset{\sim}{x}) &\stackrel{\text{def}}{=} \exp\{-\Lambda(t \mid \underset{\sim}{x})\} && \text{(survival function in terms of cumulative hazard)} \\
&= \exp\{-\theta_\lambda(\underset{\sim}{x}) \cdot \Lambda_0(t)\} && \text{(substituting cumulative hazard)} \\
&= [\exp\{-\Lambda_0(t)\}]^{\theta_\lambda(\underset{\sim}{x})} && \text{(applying exponent laws)} \\
&= [\mathrm{S}_0(t)]^{\theta_\lambda(\underset{\sim}{x})} && \text{(substituting baseline survival function } \mathrm{S}_0(t)\text{)}
\end{aligned}
$$

An alternative modeling framework is the **Accelerated Failure Time (AFT)** model. Rather than multiplying the hazard rate, covariates accelerate or decelerate the progression of time itself.

**Definition 1.2** (Acceleration factor)**.** For a covariate vector $\underset{\sim}{x}$ and coefficients $\beta_1, \ldots, \beta_p$, let $\eta(\underset{\sim}{x}) \stackrel{\text{def}}{=} \beta_1 x_1 + \cdots + \beta_p x_p$, with no intercept. The **acceleration factor** is

$$\theta_\lambda(\underset{\sim}{x}) \stackrel{\text{def}}{=} \exp\{\eta(\underset{\sim}{x})\}.$$

**Example 1.5** (Computing an acceleration factor)**.** Suppose there is one covariate, with coefficient $\beta_1 = \log\{2\}$, and a subject has $x_1 = 2$. Then:

$$
\begin{aligned}
\theta_\lambda(\underset{\sim}{x}) &= \exp\{\eta(\underset{\sim}{x})\} && \text{(definition of the acceleration factor)} \\
&= \exp\{\beta_1 x_1\} && \text{(one covariate, no intercept)} \\
&= \exp\{2\log\{2\}\} && \text{(substitute } \beta_1 = \log\{2\} \text{ and } x_1 = 2\text{)} \\
&= 4 && \text{(}\exp\{\log\{a\}\} = a\text{, and } 2\log\{2\} = \log\{4\}\text{)}
\end{aligned}
$$

**Definition 1.3** (Accelerated failure time model)**.** An **accelerated failure time (AFT) model** for an event time $T$ with covariate vector $\underset{\sim}{x}$ assumes that

$$\mathrm{S}(t \mid \underset{\sim}{x}) \stackrel{\text{def}}{=} \mathrm{S}_0(t \cdot \theta_\lambda(\underset{\sim}{x})) \tag{2}$$

where:

- $\mathrm{S}_0(t) \stackrel{\text{def}}{=} \mathbb{P}(T > t \mid \underset{\sim}{x} = \underset{\sim}{0})$ is the baseline survival function[a], and
- $\theta_\lambda(\underset{\sim}{x})$ is the acceleration factor (Definition 1.2).

[a] proportional-hazards-models.qmd#def-baseline-surv

An acceleration factor $\theta_\lambda(\underset{\sim}{x}) > 1$ makes time pass faster for a subject with covariates $\underset{\sim}{x}$ than at baseline, so events happen sooner; $\theta_\lambda(\underset{\sim}{x}) < 1$ slows time down, so events happen later.

**Example 1.6** (An acceleration factor of 4)**.** Suppose the baseline survival function is exponential with mean 40 weeks, $\mathrm{S}_0(t) = \exp\{-t/40\}$, and a covariate pattern $\underset{\sim}{x}$ has acceleration factor $\theta_\lambda(\underset{\sim}{x}) = 4$. By Equation 2, the probability that a subject with covariates $\underset{\sim}{x}$ is still event-free at 10 weeks is:

$$
\begin{aligned}
\mathrm{S}(10 \mid \underset{\sim}{x}) &= \mathrm{S}_0(10 \cdot 4) && \text{(AFT survival function with } t = 10 \text{ and } \theta_\lambda(\underset{\sim}{x}) = 4) \\
&= \mathrm{S}_0(40) && \text{(multiply)} \\
&= \exp\{-40/40\} && \text{(substitute the baseline survival function)} \\
&= \exp\{-1\} && \text{(divide)} \\
&\approx 0.368 && \text{(evaluate)}
\end{aligned}
$$

A subject with covariates $\underset{\sim}{x}$ has the same chance of lasting 10 weeks that a baseline subject has of lasting 40 weeks.

**Theorem 1.3** (AFT models rescale every quantile)**.** *In an AFT model (Definition 1.3), suppose the baseline survival function* $\mathrm{S}_0$ *is continuous and strictly decreasing, and let* $t_q^0$ *be the baseline time by which a fraction* $q$ *of subjects have had the event, so that* $\mathrm{S}_0(t_q^0) = 1 - q$*. Then the corresponding time for covariate pattern* $\underset{\sim}{x}$ *is*

$$
t_q(\underset{\sim}{x}) = \frac{t_q^0}{\theta_\lambda(\underset{\sim}{x})}.
$$

Proof

*Proof.* The time $t_q(\underset{\sim}{x})$ solves $\mathrm{S}(t_q(\underset{\sim}{x}) \mid \underset{\sim}{x}) = 1 - q$:

$$
\begin{aligned}
1 - q &= \mathrm{S}(t_q(\underset{\sim}{x}) \mid \underset{\sim}{x}) && \text{(definition of } t_q(\underset{\sim}{x})) \\
&= \mathrm{S}_0\big(t_q(\underset{\sim}{x}) \cdot \theta_\lambda(\underset{\sim}{x})\big) && \text{(AFT survival function)}
\end{aligned}
$$

Because $\mathrm{S}_0$ is strictly decreasing, $t_q^0$ is the only time at which $\mathrm{S}_0$ equals $1 - q$, so:

$$
\begin{aligned}
t_q(\underset{\sim}{x}) \cdot \theta_\lambda(\underset{\sim}{x}) &= t_q^0 && (\mathrm{S}_0 \text{ is one-to-one}) \\
t_q(\underset{\sim}{x}) &= \frac{t_q^0}{\theta_\lambda(\underset{\sim}{x})} && (\text{divide by } \theta_\lambda(\underset{\sim}{x}) > 0)
\end{aligned}
$$

□

**Example 1.7** (Median event times with an acceleration factor of 4)**.** In Example 1.6, the baseline survival function is $\mathrm{S}_0(t) = \exp\{-t/40\}$. Its median ($q = 0.5$) solves $\mathrm{S}_0(t_{0.5}^0) = 0.5$:

$$
\begin{aligned}
\exp\{-t_{0.5}^0/40\} &= 0.5 && \text{(definition of the median)} \\
-t_{0.5}^0/40 &= \log\{0.5\} && \text{(take logs)} \\
t_{0.5}^0 &= -40\log\{0.5\} && \text{(multiply by } -40) \\
&= 40\log\{2\} && (-\log\{0.5\} = \log\{1/0.5\} = \log\{2\}) \\
&\approx 27.73 && \text{(evaluate)}
\end{aligned}
$$

By Theorem 1.3, with $\theta_\lambda(\underset{\sim}{x}) = 4$, the median for covariate pattern $\underset{\sim}{x}$ is $27.73/4 \approx 6.93$ weeks. To check: $\mathrm{S}(6.93 \mid \underset{\sim}{x}) = \mathrm{S}_0(6.93 \cdot 4) = \mathrm{S}_0(27.73) = 0.5$.

We derive the expected survival time under the AFT model using change-of-variables integration:

$$
\begin{aligned}
\mathbb{E}[T \mid \underset{\sim}{x}] &\overset{\text{def}}{=} \int_0^\infty \mathrm{S}(t \mid \underset{\sim}{x}) dt && \text{(expectation formula for non-negative random variable)} \\
&= \int_0^\infty \mathrm{S}_0(t \cdot \theta_\lambda(\underset{\sim}{x})) dt && \text{(substituting AFT survival function)} \\
&= \int_0^\infty \mathrm{S}_0(u) \cdot \frac{du}{\theta_\lambda(\underset{\sim}{x})} && \text{(substituting } u = t \cdot \theta_\lambda(\underset{\sim}{x}) \implies dt = \frac{du}{\theta_\lambda(\underset{\sim}{x})}\text{)} \\
&= \theta_\lambda(\underset{\sim}{x})^{-1} \cdot \int_0^\infty \mathrm{S}_0(u) du && \text{(factoring constant } \theta_\lambda(\underset{\sim}{x})^{-1} \text{ out of integral)} \\
&= \theta_\lambda(\underset{\sim}{x})^{-1} \cdot \mathbb{E}[T \mid \underset{\sim}{x} = \underset{\sim}{0}] && \text{(substituting baseline expectation formula)}
\end{aligned}
$$

Thus, the mean survival time for a subject with covariates $\underset{\sim}{x}$ equals the baseline mean survival time divided by the acceleration factor $\theta_\lambda(\underset{\sim}{x}) = \exp\{\eta(\underset{\sim}{x})\}$.

We also derive the cumulative hazard and hazard rate functions under the AFT formulation:

$$
\begin{aligned}
\Lambda(t \mid \underset{\sim}{x}) &\overset{\text{def}}{=} -\log\{\mathrm{S}(t \mid \underset{\sim}{x})\} && \text{(cumulative hazard function definition)} \\
&= -\log\{\mathrm{S}_0(t \cdot \theta_\lambda(\underset{\sim}{x}))\} && \text{(substituting AFT survival function)} \\
&= \Lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x})) && \text{(substituting baseline cumulative hazard)} \\
\lambda(t \mid \underset{\sim}{x}) &\overset{\text{def}}{=} \frac{\partial}{\partial t} \Lambda(t \mid \underset{\sim}{x}) && \text{(hazard function definition)} \\
&= \frac{\partial}{\partial t} (\Lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x}))) && \text{(substituting AFT cumulative hazard)} \\
&= \Lambda_0'(t \cdot \theta_\lambda(\underset{\sim}{x})) \cdot \frac{\partial}{\partial t}(t \cdot \theta_\lambda(\underset{\sim}{x})) && \text{(applying chain rule of differentiation)} \\
&= \lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x})) \cdot \theta_\lambda(\underset{\sim}{x}) && \text{(since } \Lambda_0'(u) = \lambda_0(u)\text{)} \\
&= \theta_\lambda(\underset{\sim}{x}) \cdot \lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x})) && \text{(rearranging factors)}
\end{aligned}
$$

### 1.4.1 Relationship between PH and AFT models

If the baseline distribution is exponential with parameter $\lambda$, substituting $\mathrm{S}_0(t) = \exp\{-\lambda t\}$ into the AFT model yields:

$$
\begin{aligned}
\mathrm{S}(t \mid \underset{\sim}{x}) &= \exp\{-\lambda \cdot t \cdot \theta_\lambda(\underset{\sim}{x})\} && \text{(substituting exponential baseline into AFT formula)} \\
&= [\exp\{-\lambda t\}]^{\theta_\lambda(\underset{\sim}{x})} && \text{(applying exponent laws)} \\
&= [\mathrm{S}_0(t)]^{\theta_\lambda(\underset{\sim}{x})} && \text{(substituting exponential baseline survival function)}
\end{aligned}
$$

This derivation demonstrates that the exponential AFT model is mathematically identical to the exponential PH model.

The Weibull distribution has the same property.

**Theorem 1.4** (A Weibull proportional hazards model is an AFT model)**.** *Let* $\mathrm{S}_0(t) = \exp\{-(\lambda t)^p\}$ *be a Weibull baseline survival function with scale* $\lambda > 0$ *and shape* $p > 0$*. If subject* $i$ *has hazard* $\lambda_0(t) \cdot \theta_i$ *for a constant* $\theta_i > 0$*, then subject* $i$*'s survival function is*

$$\mathrm{S}_i(t) = \mathrm{S}_0\big(t \cdot \theta_i^{1/p}\big),$$

*so the model is also an accelerated failure time model, with acceleration factor* $\theta_i^{1/p}$*.*

Proof

*Proof.*

$$
\begin{aligned}
\mathrm{S}_i(t) &\overset{\text{def}}{=} \exp\left\{-\int_0^t \lambda_0(u)\cdot\theta_i\,du\right\} && \text{(survival function in terms of the hazard)}\\
&= \exp\{-\theta_i\cdot\Lambda_0(t)\} && \text{(factor the constant } \theta_i \text{ out of the integral)}\\
&= \exp\{-\theta_i(\lambda t)^p\} && \text{(Weibull cumulative hazard } \Lambda_0(t)=(\lambda t)^p)\\
&= \exp\left\{-\left(\theta_i^{1/p}\right)^p(\lambda t)^p\right\} && \text{(write } \theta_i=\left(\theta_i^{1/p}\right)^p)\\
&= \exp\left\{-\left(\lambda\cdot t\cdot\theta_i^{1/p}\right)^p\right\} && \text{(combine the } p\text{-th powers)}\\
&= \mathrm{S}_0\left(t\cdot\theta_i^{1/p}\right) && \text{(Weibull baseline survival function at } t\cdot\theta_i^{1/p})
\end{aligned}
$$

□

**Example 1.8** (A hazard ratio of 4 with Weibull shape 2)**.** Take $\lambda = 0.05$, $p = 2$, and $\theta_i = 4$, with time in weeks. By Theorem 1.4, the acceleration factor is

$$
\begin{aligned}
\theta_i^{1/p} &= 4^{1/2} && \text{(substitute } \theta_i=4,\ p=2)\\
&= 2 && \text{(square root)}
\end{aligned}
$$

To check, compute subject $i$'s survival at 10 weeks both ways:

$$
\begin{aligned}
[\mathrm{S}_0(10)]^{\theta_i} &= [\exp\{-(0.05\cdot 10)^2\}]^4 && \text{(proportional hazards form)}\\
&= \exp\{-4\cdot 0.25\} && ((0.5)^2=0.25\text{; exponent laws})\\
&= \exp\{-1\} && \text{(multiply)}\\
\mathrm{S}_0(10\cdot 2) &= \exp\{-(0.05\cdot 20)^2\} && \text{(AFT form with acceleration factor 2)}\\
&= \exp\{-1\} && ((1)^2=1)
\end{aligned}
$$

Both forms give $\exp\{-1\} \approx 0.368$: a four-fold hazard means that subject $i$ ages twice as fast as baseline.

**Theorem 1.5** (The Weibull model is a log-linear model for event times)**.** *Suppose subject $i$ has the Weibull proportional hazards survival function $\mathrm{S}_i(t) = \exp\{-\theta_i(\lambda t)^p\}$ of Theorem 1.4, with $\theta_i \overset{def}{=} \exp\{\eta_i\}$ for a linear predictor $\eta_i$. Define $\alpha \overset{def}{=} -\log\{\lambda\}$ and $\sigma \overset{def}{=} 1/p$, and let $W$ have the standard (minimum) Gumbel distribution, $\mathbb{P}(W > w) = \exp\{-e^w\}$. Then the log event time $Y_i \overset{def}{=} \log\{T_i\}$ has the same distribution as*

$$\alpha - \sigma\eta_i + \sigma W. \tag{3}$$

Proof

*Proof.* For any real $y$:

$$
\begin{aligned}
\mathbb{P}(Y_i > y) &= \mathbb{P}(T_i > \mathrm{e}^y) && (\log \text{ is increasing}) \\
&= \exp\{-\theta_i(\lambda \mathrm{e}^y)^p\} && (\text{Weibull PH survival function at } \mathrm{e}^y) \\
&= \exp\{-\mathrm{e}^{\eta_i}(\lambda \mathrm{e}^y)^p\} && (\theta_i = \mathrm{e}^{\eta_i}) \\
&= \exp\{-\mathrm{e}^{\eta_i}\lambda^p \mathrm{e}^{py}\} && (\text{power of a product}) \\
&= \exp\left\{-\mathrm{e}^{\eta_i}\mathrm{e}^{p\log\{\lambda\}}\mathrm{e}^{py}\right\} && (\lambda^p = \mathrm{e}^{p\log\{\lambda\}}) \\
&= \exp\left\{-\mathrm{e}^{\eta_i + p\log\{\lambda\} + py}\right\} && (\text{product rule for exponentials}) \\
&= \exp\left\{-\mathrm{e}^{\eta_i + (y-\alpha)/\sigma}\right\} && (\text{substitute } p = 1/\sigma,\ \log\{\lambda\} = -\alpha) \\
&= \exp\left\{-\mathrm{e}^{(y - \alpha + \sigma\eta_i)/\sigma}\right\} && (\text{common denominator } \sigma) \\
&= \mathbb{P}\left(W > \frac{y - \alpha + \sigma\eta_i}{\sigma}\right) && (\text{Gumbel survival function}) \\
&= \mathbb{P}(\sigma W > y - \alpha + \sigma\eta_i) && (\text{multiply both sides by } \sigma > 0) \\
&= \mathbb{P}(\alpha - \sigma\eta_i + \sigma W > y) && (\text{add } \alpha - \sigma\eta_i \text{ to both sides})
\end{aligned}
$$

The two random variables have the same survival function, so they have the same distribution. □

**Example 1.9** (Median event times from the log-linear form)**.** Take $\lambda = 0.1$ and $p = 2$, so $\alpha = -\log\{0.1\} = \log\{10\} \approx 2.3026$ and $\sigma = 0.5$. The median of $W$ solves $\exp\{-\mathrm{e}^w\} = 0.5$, so $\mathrm{e}^w = \log\{2\}$ and $w = \log\{\log\{2\}\} \approx -0.3665$. Because $\alpha - \sigma\eta + \sigma W$ is an increasing function of $W$, the median of $Y$ is that function evaluated at the median of $W$.
For a baseline subject ($\eta = 0$):

$$
\begin{aligned}
\text{median}(Y) &= \alpha + \sigma \cdot (-0.3665) && (\text{log-linear form at the median of } W) \\
&= 2.3026 - 0.1833 && (\text{multiply}) \\
&= 2.1193 && (\text{subtract})
\end{aligned}
$$

so the median event time is $\mathrm{e}^{2.1193} \approx 8.33$. Directly, $\mathrm{S}_0(8.33) = \exp\{-(0.833)^2\} = \exp\{-0.694\} \approx 0.5$.
For a subject with hazard ratio $\theta = 4$, so $\eta = \log\{4\} \approx 1.3863$, the median of $Y$ shifts by $-\sigma\eta = -0.5 \cdot 1.3863 \approx -0.6931$, so the median event time is multiplied by $\mathrm{e}^{-0.6931} = 0.5$, giving $8.33 \cdot 0.5 \approx 4.16$.

By Equation 3, the estimated AFT regression coefficients $\hat{\beta}_{\text{AFT}}$ produced by R's `survreg()` function relate directly to the PH regression coefficients $\hat{\beta}_{\text{PH}}$ produced by `coxph()` via the identity $\hat{\beta}_{\text{AFT}} = -\sigma\hat{\beta}_{\text{PH}}$.

The exponential and Weibull distributions are the only continuous distributions that are simultaneously proportional hazards models and accelerated failure time models. Other parametric families commonly used for survival data (log-logistic, log-normal, and generalized gamma) are AFT families rather than PH families; the log-logistic distribution is additionally a proportional-odds model.

## 1.5 Dataset: Leukemia treatments

To illustrate semi-parametric and parametric survival models, we analyze remission survival times from a clinical trial of 42 pediatric leukemia patients [2]. Half of the patients received a new therapy (6-mercaptopurine, 6-MP) and half received a standard control treatment.

The variables in this dataset include:

- `survt`: Remission duration in weeks ($T$).
- `status`: Relapse status (`relapse` = 1 or `censored` = 0).
- `rx`: Treatment assignment (`new` vs `standard`).
- `sex`: Patient sex (`female` vs `male`).
- `surv`: Survival outcome object created with `Surv(time = survt, event = (status == "relapse"))`.

```r
library(haven)
library(survival)
anderson <-
  paste0(
    "http://web1.sph.emory.edu/dkleinb/allDatasets",
    "/surv2datasets/anderson.dta"
  ) |>
  read_dta() |>
  mutate(
    status = status |>
      case_match(
        1 ~ "relapse",
        0 ~ "censored"
      ),
    sex = sex |>
      case_match(
        0 ~ "female",
        1 ~ "male"
      ),

    rx = rx |>
      case_match(
        0 ~ "new",
        1 ~ "standard"
      ),

    surv = Surv(time = survt, event = (status == "relapse"))
  )

print(anderson)
```

### 1.5.1 Cox semi-parametric model

We fit a Cox proportional hazards model[14] comparing standard therapy to new therapy:

```r
anderson_cox0 <- coxph(
  formula = surv ~ rx,
  data = anderson
)
summary(anderson_cox0)
#> Call:
#> coxph(formula = surv ~ rx, data = anderson)
#>
#>   n= 42, number of events= 30
#>
#>             coef exp(coef) se(coef)    z Pr(>|z|)
#> rxstandard 1.572     4.817    0.412 3.81  0.00014 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#>            exp(coef) exp(-coef) lower .95 upper .95
#> rxstandard      4.82      0.208      2.15      10.8
#>
#> Concordance= 0.69  (se = 0.041 )
#> Likelihood ratio test= 16.4  on 1 df,   p=5e-05
#> Wald test            = 14.5  on 1 df,   p=1e-04
```

[14] proportional-hazards-models.qmd#def-ph-model

```
#> Score (logrank) test = 17.2  on 1 df,   p=3e-05
```

The estimated log hazard ratio[15] $\hat{\beta}_{\text{PH}}$ for standard treatment relative to new treatment is positive, indicating that standard therapy is associated with a higher hazard of relapse (shorter remission duration).

### 1.5.2 Weibull parametric model

We next fit a parametric Weibull accelerated failure time model using `survreg()`:

```
anderson_weib <- survreg(
  formula = surv ~ rx,
  data = anderson,
  dist = "weibull"
)
summary(anderson_weib)
#>
#> Call:
#> survreg(formula = surv ~ rx, data = anderson, dist = "weibull")
#>              Value Std. Error     z       p
#> (Intercept)  3.516      0.252 13.96 < 2e-16
#> rxstandard  -1.267      0.311 -4.08 4.5e-05
#> Log(scale)  -0.312      0.147 -2.12   0.034
#>
#> Scale= 0.732
#>
#> Weibull distribution
#> Loglik(model)= -106.6   Loglik(intercept only)= -116.4
#>  Chisq= 19.65 on 1 degrees of freedom, p= 9.3e-06
#> Number of Newton-Raphson Iterations: 5
#> n= 42
```

In `survreg()`, regression parameters are reported on the log-time scale ($Y = \log T = \alpha + \underset{\sim}{\beta}_{\text{AFT}}\underset{\sim}{x} + \sigma W$). The estimated coefficient $\hat{\beta}_{\text{AFT}}$ for standard treatment is negative, indicating a reduction in log survival time (faster progression to relapse). The scale estimate $\hat{\sigma}$ corresponds to $1/p$, and satisfies $\hat{\beta}_{\text{AFT}} = -\hat{\sigma}\hat{\beta}_{\text{PH}}$.

### 1.5.3 Exponential parametric model

Setting the scale parameter to $\sigma = 1$ ($p = 1$) yields the exponential parametric model:

```
anderson_exp <- survreg(
  formula = surv ~ rx,
  data = anderson,
  dist = "exp"
)
summary(anderson_exp)
#>
#> Call:
#> survreg(formula = surv ~ rx, data = anderson, dist = "exp")
#>              Value Std. Error     z       p
#> (Intercept)  3.686      0.333 11.06 < 2e-16
#> rxstandard  -1.527      0.398 -3.83 0.00013
#>
#> Scale fixed at 1
#>
#> Exponential distribution
#> Loglik(model)= -108.5   Loglik(intercept only)= -116.8
```

[15] intro-to-survival-analysis.qmd#def-hazard-ratio

```
#>  Chisq= 16.49 on 1 degrees of freedom, p= 4.9e-05
#> Number of Newton-Raphson Iterations: 4
#> n= 42
```

In the exponential model, $\hat{\beta}_{\text{AFT}} = -\hat{\beta}_{\text{PH}}$, so the AFT coefficient is equal in magnitude and opposite in sign to the Cox proportional hazards coefficient.

**Example 1.10** (Exponential rate estimates for the leukemia trial)**.** By the MLE of the exponential rate parameter[a], each group's estimated hazard rate is its number of relapses divided by its total follow-up time:

```
anderson |>
  summarize(
    .by = rx,
    relapses = sum(status == "relapse"),
    follow_up = sum(survt),
    rate = relapses / follow_up
  )
#> # A tibble: 2 x 4
#>   rx       relapses follow_up   rate
#>   <chr>       <int>     <dbl>  <dbl>
#> 1 new             9       359 0.0251
#> 2 standard       21       182 0.115
```

For the new treatment, 9 relapses over 359 weeks of follow-up give $\hat{\lambda}_{\text{new}} = 9/359 \approx 0.0251$ relapses per week; for the standard treatment, 21 relapses over 182 weeks give $\hat{\lambda}_{\text{standard}} = 21/182 \approx 0.1154$. The estimated log hazard ratio is then:

$$\begin{aligned}
\log\left\{\frac{\hat{\lambda}_{\text{standard}}}{\hat{\lambda}_{\text{new}}}\right\} &= \log\left\{\frac{21/182}{9/359}\right\} && \text{(substitute the rate estimates)} \\
&= \log\left\{\frac{21 \cdot 359}{182 \cdot 9}\right\} && \text{(divide fractions)} \\
&= \log\left\{\frac{7539}{1638}\right\} && \text{(multiply)} \\
&\approx 1.527 && \text{(evaluate)}
\end{aligned}$$

The `survreg()` fit's `rxstandard` coefficient, $-1.527$, is the negative of this log hazard ratio, as the exponential AFT model predicts, and its intercept, 3.686, is $\log\{359/9\}$, the log of the estimated mean remission time on the new treatment.

[a] intro-to-survival-analysis.qmd#thm-mle-exp

### 1.5.4 Diagnostic: complementary log-log survival plot

To check whether a Weibull proportional hazards assumption[16] is reasonable, we inspect the complementary log-log transformation of the Kaplan-Meier survival curves[17], $\log(-\log \hat{\text{S}}(t)) = p \log \lambda + p \log t$:

```
# plot the cumulative hazard H(t) = -log(S(t)) against time on log-spaced axes,
# so that the ticks are in weeks and in units of cumulative hazard,
# rather than in the logs of those quantities;
# log10 is not finite at 0, so leave out the points where H(t) = 0
# (before the first event) and where S(t) = 0 (so that H(t) is infinite):
survfit(
  formula = surv ~ rx,
  data = anderson
) |>
```

[16] proportional-hazards-models.qmd#def-pha
[17] intro-to-survival-analysis.qmd#def-KM-estimator

```r
broom::tidy() |>
dplyr::mutate(cumhaz = -log(estimate)) |>
dplyr::filter(time > 0, cumhaz > 0, is.finite(cumhaz)) |>
ggplot() +
aes(x = time, y = cumhaz, col = strata) +
geom_step() +
theme_bw() +
scale_x_continuous(
  trans = "log10",
  labels = scales::label_number(drop0trailing = TRUE, big.mark = ","),
  name = "Time (weeks, log scale)"
) +
scale_y_continuous(
  trans = "log10",
  labels = scales::label_number(drop0trailing = TRUE, big.mark = ","),
  name = "Cumulative hazard H(t) (log scale)"
) +
theme(legend.position = "bottom", legend.title = element_blank())
```

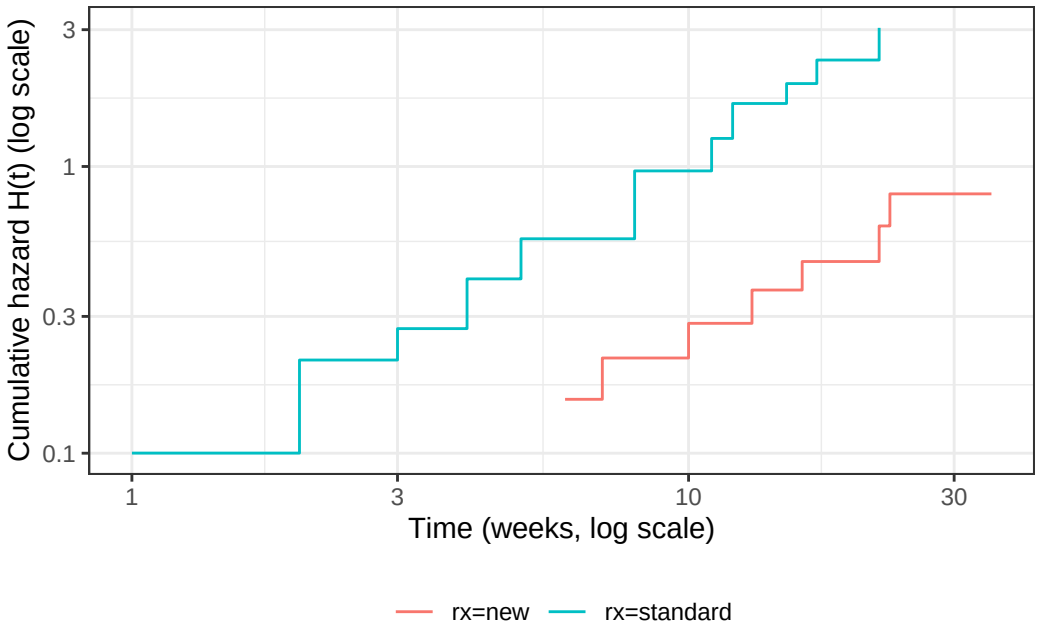



Figure 4: Cumulative hazard curves by treatment group, on log-spaced axes (complementary log-log survival curves)

If the complementary log-log curves are approximately linear and parallel across treatment groups, then a Weibull proportional hazards model is appropriate.

## 2 Combining left-truncation and interval-censoring

In observational studies and clinical trials, complex observation schemes can introduce both **left-truncation**[18] (delayed entry) and **interval-censoring**[19]. Left-truncation occurs when subjects enter the risk set[20] only after surviving past an entry time $L_{\text{entry}} > 0$. Interval-censoring occurs when the exact event time $T_i$ is unknown, but is known to fall within an interval $(L_i, R_i]$.

[18] intro-to-survival-analysis.qmd#def-left-truncation
[19] intro-to-survival-analysis.qmd#def-interval-censoring
[20] intro-to-survival-analysis.qmd#def-risk-set

Standard software tools in R make distinct trade-offs between these two observation mechanisms:

> `coxph` does left truncation but not left (or interval) censoring; `survreg` does interval censoring but not left truncation (or time dependent covariates).
>
> — Terry Therneau, R-help, August 31, 2015[21]

In particular, `coxph()` constructs risk sets dynamically at each observed failure time $t_j$ using counting process format (`time1 = entry`, `time2 = exit`), enabling seamless handling of left-truncation. However, because partial likelihood[22] relies on ordered exact failure times, standard Cox models cannot easily accommodate interval-censored data.

Conversely, `survreg()` accommodates interval-censored outcomes via `Surv(time1 = L, time2 = R, type = "interval")` by maximizing the interval parametric likelihood[23] $\mathbb{P}(L_i < T_i \leq R_i) = \mathrm{F}_0(R_i) - \mathrm{F}_0(L_i)$. However, standard `survreg()` assumes all subjects enter observation at time 0, and does not condition likelihood terms on surviving past a delayed entry time $L_{\text{entry}}$.

When a study contains both delayed entry and interval-censored event times, specialized tooling is needed. Among the following R packages, only `flexsurv` handles both features together; the other two are listed because they cover interval censoring well and are the usual starting points, not because they solve the truncation half:

- `flexsurv`: Fits flexible parametric survival models and allows custom likelihood specifications that incorporate both left-truncation conditioning and interval-censored bounds.
- `icenReg`: Fits regression models for interval-censored data: Cox proportional-hazards, proportional-odds, and accelerated failure time. Semi-parametric and fully parametric forms are both available, but the fully parametric option covers the AFT models only. It does not itself provide left-truncation conditioning.
- `interval`: Fits nonparametric survival curves (NPMLE) for interval-censored data and provides weighted logrank and Wilcoxon-type tests [3]. It is nonparametric throughout, so it supports neither regression nor truncation.


[1] D. M. Rocke, "Epi 204 lecture notes, Spring 2021." University of California, Davis, 2021. Available: https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/EPI204-Spring-2021.html

[2] J. D. Kalbfleisch and R. L. Prentice, *The statistical analysis of failure time data*. John Wiley & Sons, 2011.

[3] M. P. Fay and P. A. Shaw, "Exact and asymptotic weighted logrank tests for interval censored data: The interval R package," *Journal of Statistical Software*, vol. 36, no. 2, pp. 1–34, 2010, doi: 10.18637/jss.v036.i02[24].


[21] https://stat.ethz.ch/pipermail/r-help/2015-August/431733.html
[22] proportional-hazards-models.qmd#def-ph-partial-lik
[23] https://morrison-lab.github.io/sds/intro-MLEs.html#def-lik
[24] https://doi.org/10.18637/jss.v036.i02